Quiz 7 — Instructor Key

Friday, October 30, 2026 10 points

1. (2 points) By the product rule,

$$v'(x) = e^{2x} + 2xe^{2x} = e^{2x}(1 + 2x).$$

Since $e^{2x} > 0$ for every real x, $v'(x) = 0$ exactly at $x = -\frac{1}{2}$. The derivative exists everywhere, so this is the only critical number. The corresponding point is $(-\frac{1}{2}, -\frac{1}{2e})$. Only the x-coordinates are requested; no classification is required in Problem 1.

2. (3 points)

$$u'(x) = 12x^2 + 18x - 12 = 6(2x - 1)(x + 2).$$

The derivative exists for every real x and vanishes exactly at $x = -2$ and $x = \frac{1}{2}$. It is positive on $(-\infty, -2)$, negative on $(-2, \frac{1}{2})$, and positive on $(\frac{1}{2}, \infty)$. Thus $x = -2$ gives a local maximum $(+ \to -)$, and $x = \frac{1}{2}$ gives a local minimum $(- \to +)$. For reference, the corresponding points are $(-2, 33)$ and $(\frac{1}{2}, \frac{7}{4})$.

3. (2 points) $x = -2$: local maximum; $x = 0$: local minimum; $x = 2$: saddle point (stationary inflection point). At -2, f changes from increasing to decreasing. At 0, the sharp corner is a local minimum: f decreases to the corner and increases after it. At 2, the tangent is horizontal, but f is increasing on both sides, and its concavity changes there, giving a stationary inflection point. There are no other critical numbers in $(-4, 4)$.

4. (2 points) (a) The zeros of f' are -3, -1, 1, and 3. Its sign chart is

x	$(-4,-3)$	$(-3,-1)$	$(-1,1)$	$(1,3)$	$(3,4)$
$f'(x)$	$-$	$+$	$-$	$-$	$+$

(b) Since f is differentiable throughout $(-4, 4)$, its critical numbers are exactly these zeros: -3, -1, 1, and 3. At $x = -3$, f has a local minimum $(- \to +)$; at $x = -1$, a local maximum $(+ \to -)$; at $x = 1$, a saddle point (stationary inflection point); and at $x = 3$, a local minimum $(- \to +)$. At 1, f' is negative on both sides and has a local maximum, so f changes concavity there without having an extremum. At 0, the derivative graph has a corner and a local minimum, but $f'(0) = -1$ is defined and nonzero, so 0 is not a critical number of f.

It is smooth except for the corner at 0, where its left and right slopes are -2 and 2, making 0 a local minimum of f'. The zero at -1 is simple: the slope of f' there is $-\frac{4}{3} \neq 0$. The other displayed extrema of f' are a local maximum at -2 (height 1), a local maximum at 1 (height 0), and a local minimum at 2 (height $-\frac{1}{2}$).

5. (1 point) (a) False. (b) False. (c) True. (d) False.

Grading: Problem 5 requires only the correct circled choices; no explanation is required. Give partial credit for correct rule setup or sign analysis, and carry minor arithmetic errors forward without penalizing the same error again. In Problem 3, do not count the endpoints: the question is restricted to the open interval. The same applies to Problem 4.